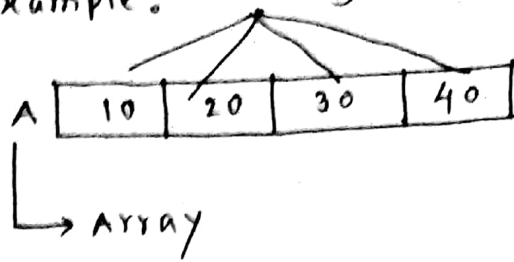


Data structures:

A particular way of organizing data in a computer so that it can be used effectively.

For example: integer data



We can store a list of items having the same data-type using the array data structure.

Different Data structures

- Linear: arrays, lists
- Tree: Binary, heaps, space partitioning, etc
- stack
- Queue
- Graph
- Files
- Records

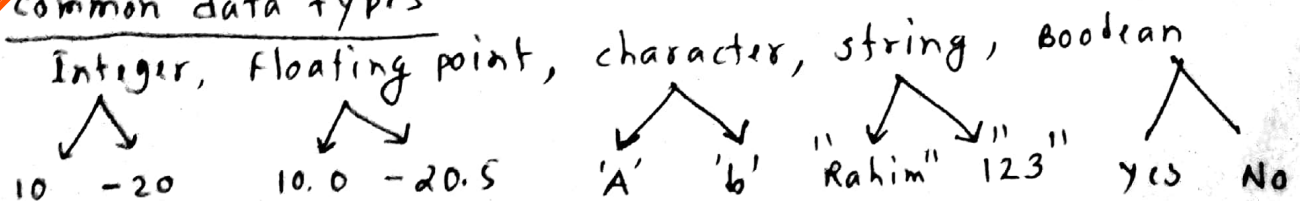
Uses of Data structures

- Operating system
- Compiler design
- Artificial Intelligence
- Graphics

Fundamental data structures

- Array
- Linked list

Common data types



What is algorithms?

step by step problem solution technique

Algorithm Representation

- Pseudocode
- flowchart
- Program

Array Data structure

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Sorting:

A

10	20	30
----	----	----

0 1 2

Ascending order

A

30	20	10
----	----	----

0 1 2

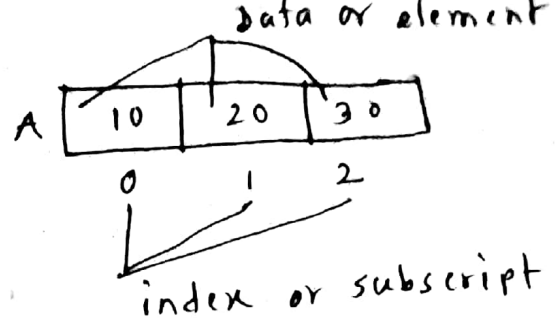
Descending order

Name

Abu	Babu	Robi
-----	------	------

0 1 2

Alphabetical order



$A[0] = 10$

$A[1] = 20$

$A[2] = 30$

A

9	8	6	3	3	1
---	---	---	---	---	---

0 1 2 3 4 5

Non-increasing order

A

1	3	3	6	8	9
---	---	---	---	---	---

0 1 2 3 4 5

Non-decreasing order

Different types of sorting Techniques

- Bubble Sort
- Selection Sort
- Merge Sort *
- Insertion Sort *
- Quick Sort *
- Heap Sort *

- Counting Sort *
- Radix Sort *
- Bucket Sort

Linear
time
sorting

Different cases of
sorting: (sort in ascending
order)

Best case: A

10	20	30
----	----	----

0 1 2

Worst case: A

30	20	10
----	----	----

0 1 2

Average case: A

10	30	20
----	----	----

0 1 2

Counting Sort

Sort the Array A in ascending order using counting sort

A	4	2	3	2	4	1
	1	2	3	4	5	6

Ans:

Input Data: A

4	2	3	2	4	1
1	2	3	4	5	6

Find Max number: $k=4$

Initialize Mapping Array: C

0	0	0	0	0	0
0	1	2	3	4=k	

count number stored in Array A: C

0	1	2	1	2	
0	1	2	3	4	

find Proper Mapping Array: C

0	1	3	4	6	
0	1	2	3	4	

$j=6$ $A[6]=1$ $C[1]=1$ $B[1]=1$

1					
1	2	3	4	5	6

 $C[1]=0$ C

0	0	3	4	6	
0	1	2	3	4	

$j=5$ $A[5]=4$ $C[4]=6$ $B[6]=4$

1				4	
1	2	3	4	5	6

 $C[4]=5$ C

0	0	3	4	5	
0	1	2	3	4	

$j=4$ $A[4]=2$ $C[2]=3$ $B[3]=2$

1	2			4	
1	2	3	4	5	6

 $C[2]=2$ C

0	0	2	4	5	
0	1	2	3	4	

$j=3$ $A[3]=3$ $C[3]=4$ $B[4]=3$

1	2	3		4	
1	2	3	4	5	6

 $C[3]=3$ C

0	0	2	3	5	
0	1	2	3	4	

$j=2$ $A[2]=2$ $C[2]=2$ $B[2]=2$

1	2	2	3		4
1	2	3	4	5	6

 $C[2]=1$ C

0	0	1	3	5	
0	1	2	3	4	

$j=1$ $A[1]=4$ $C[4]=5$ $B[5]=4$

1	2	2	3	4	4
1	2	3	4	5	6

 $C[4]=4$ C

0	0	1	3	4	
0	1	2	3	4	

Algorithm

countingsort(A, B, k) { $\Rightarrow$ output

for $i = 0$ to k

$C[i] = 0$

for $j = 1$ to n

$C[A[j]] = C[A[j]] + 1$

for $i = 2$ to k

$C[i] = C[i] + C[i-1]$

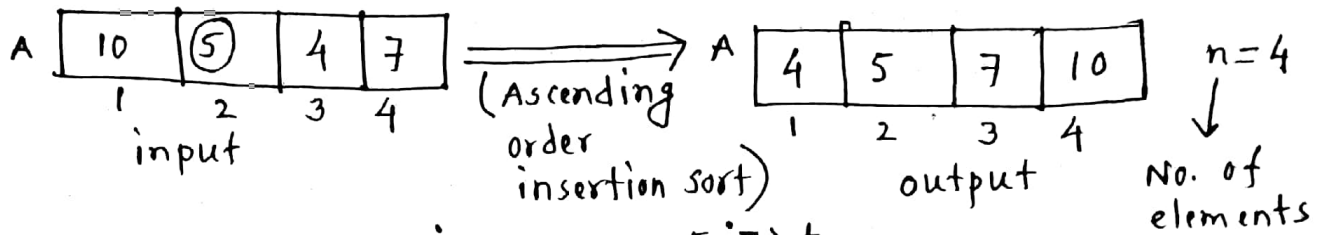
for $j = n$ down to 1

$B[C[A[j]]] = A[j]$

$C[A[j]] = C[A[j]] - 1$

Insertion Sort

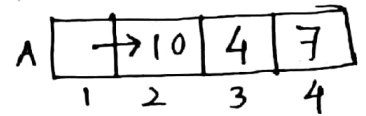
Show the mechanism of insertion sort for the following data



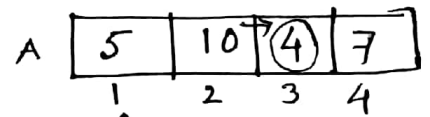
Ans:

$$j=2 \quad t=A[2] \quad i=j-1 \\ =5 \quad =2-1 \\ =1$$

$$i \geq 1 \text{ AND } A[i] > t \\ 1 \geq 1 \text{ AND } A[1] > 5 \\ \Rightarrow \text{True AND } 10 > 5 = \text{True}$$

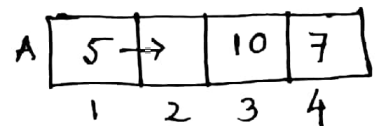


$$i=0 \quad 0 \geq 1 \text{ AND } - \\ \Rightarrow \text{False AND } - \\ i+1 = 0+1 = 1$$

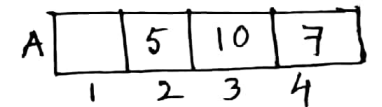


$$j=3 \quad t=A[3] \quad i=j-1 \\ =4 \quad =3-1 \\ =2$$

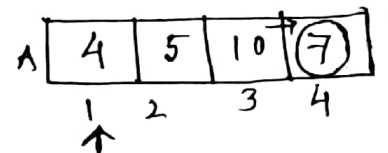
$$2 \geq 1 \text{ AND } A[2] > 4 \\ \Rightarrow \text{True AND } 10 > 4 = \text{True}$$



$$i=1 \quad 1 \geq 1 \text{ AND } A[1] > 4 \\ \Rightarrow \text{True AND } 5 > 4 = \text{True}$$

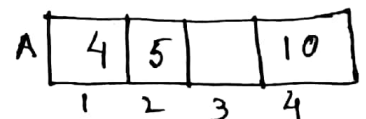


$$i=0 \quad 0 \geq 1 \text{ AND } - = \text{False} \\ \text{False} \quad i+1 = 0+1 = 1$$

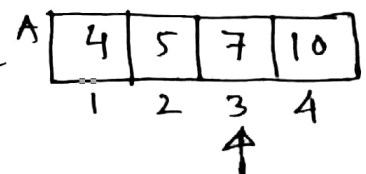


$$j=4 \quad t=A[4] \quad i=j-1 \\ \downarrow n \quad =7 \quad =4-1 \\ =3$$

$$3 \geq 1 \text{ AND } A[3] > 7 \\ \Rightarrow \text{True AND } 10 > 7 = \text{True}$$



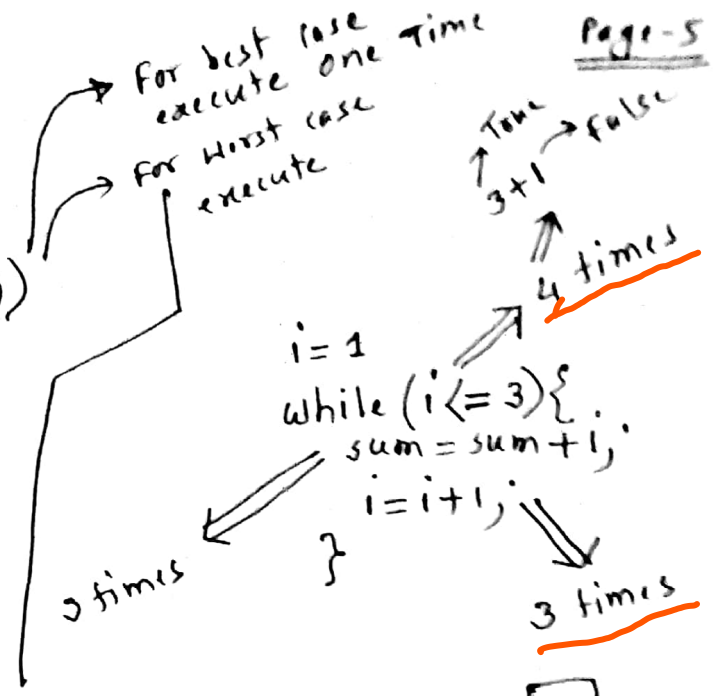
$$i=2 \quad 2 \geq 1 \text{ AND } A[2] > 7 \\ \Rightarrow \text{True AND } 5 > 7 = \text{False} \\ i+1 = 2+1 = 3$$



Algorithm

```

L1: for j = 2 to n
L2:   t = A[j]
L3:   i = j-1
L4:   while ((i >= 1) AND (A[i] > t))
L5:     A[i+1] = A[i]
L6:     i = i-1
      end while
L7:   A[i+1] = t
end for
  
```



Time Complexity Analysis

Best case Analysis:

A

10	20	30
1	2	3



Line Number	Iteration	Time/Iteration	Time
L1	$(n-2+1)+1$ True False	c_1	$c_1 n$
L2	$n-1$	c_2	$c_2(n-1)$
L3	$n-1$	c_3	$c_3(n-1)$
L4	$n-1$	c_4	$c_4(n-1)$
L5	X	X	X
L6	X	X	X
L7	$n-1$	c_7	$c_7(n-1)$

Total running time, $T(n) = c_1 n + c_2(n-1) + c_3(n-1) + c_4(n-1) + c_7(n-1)$

$= (c_1 + c_2 + c_3 + c_4 + c_7) n + (-c_2 - c_3 - c_4 - c_7)$

$= An + B = O(n)$

Worst Case Analysis

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A	30	20	10
	1	2	3

$j=2$ $t=20$ $i=1$ True } 2 times
 $i=0$ False } $\downarrow j$

$j=3$ $t=10$ $i=2$ True } 3 times
 $i=1$ True } $\downarrow j$
 $i=0$ False } j times

Line Number	Iteration	Time/Iteration	Time
L1	$(n-2+1) + \underbrace{1}_{\text{True False}} = n$	c_1	$c_1 n$
L2	$n-1$	c_2	$c_2 (n-1)$
L3	$n-1$	c_3	$c_3 (n-1)$
L4	$\sum_{j=2}^n j$	c_4	$c_4 \sum_{j=2}^n j$
L5	$\sum_{j=2}^n (j-1)$	c_5	$c_5 \sum_{j=2}^n (j-1)$
L6	$\sum_{j=2}^n (j-1)$	c_6	$c_6 \sum_{j=2}^n (j-1)$
L7	$n-1$	c_7	$c_7 (n-1)$

Total running time, $T(n) = c_1 n + c_2 (n-1) + c_3 (n-1) + c_4 \sum_{j=2}^n j$
 $+ c_5 \sum_{j=2}^n (j-1) + c_6 \sum_{j=2}^n (j-1) + c_7 (n-1)$

$$\begin{pmatrix} n \\ 2 \end{pmatrix} \sqrt{\frac{n}{2}} \quad \begin{pmatrix} n \\ 2 \end{pmatrix} \sqrt{\frac{n}{2}}$$

$$\begin{matrix} n \\ + \\ n \\ + \\ \dots \\ + \\ 1+2+\dots+n \end{matrix} \quad \begin{matrix} n \\ + \\ n-1 \\ + \\ n-2 \\ + \\ \dots \\ + \\ 1 \end{matrix}$$

$$= c_1 n + c_2 (n-1) + c_3 (n-1) + c_4 (2+3+\dots+n)$$

$$+ c_5 (1+2+\dots+n-1) + c_6 (1+2+3+\dots+n-1)$$

$$+ c_7 (n-1)$$

$$= c_1 n + c_2 (n-1) + c_3 (n-1) + c_4 (1+2+3+\dots+n-1)$$

$$+ c_5 \frac{n(n-1)}{2} + c_6 \frac{n(n-1)}{2} + c_7 (n-1)$$

$$= c_1 n + c_2(n-1) + c_3(n-1) + c_4 \left\{ \frac{n(n+1)}{2} - 1 \right\} + c_5 \frac{n(n-1)}{2} + c_6 \frac{n(n-1)}{2} + c_7(n-1)$$

$$= c_1 n + c_2 n - c_2 + c_3 n - c_3 + \frac{c_4}{2} n^2 + \frac{c_4}{2} n - c_4 + \frac{c_5}{2} n^2 - \frac{c_5}{2} n + \frac{c_6}{2} n^2 - \frac{c_6}{2} n + c_7 n - c_7$$

$$= \left(\frac{c_4}{2} + \frac{c_5}{2} + \frac{c_6}{2} \right) n^2 + \left(c_1 + c_2 + c_3 + \frac{c_4}{2} - \frac{c_5}{2} - \frac{c_6}{2} + c_7 \right) n + (-c_2 - c_3 - c_4 - c_7)$$

$$= An^2 + Bn + C$$

$$= O(n^2)$$

Average case Analysis

$$\text{condition of while loop} = \frac{\text{best} + \text{Worst}}{2} = \frac{1+j}{2}$$

<u>Line Number</u>	<u>Iteration</u>	<u>Time/Iteration</u>	<u>Time</u>
L1	n	c ₁	c ₁ n
L2	n-1	c ₂	c ₂ (n-1)
L3	n-1	c ₃	c ₃ (n-1)
L4	$\sum_{j=2}^n \frac{1+j}{2}$	c ₄	$c_4 \sum_{j=2}^n \frac{1+j}{2}$
L5	$\sum_{j=2}^n \left\{ \frac{1+j}{2} - 1 \right\}$	c ₅	$c_5 \sum_{j=2}^n \frac{j-1}{2}$
L6	$\sum_{j=2}^n \left\{ \frac{1+j}{2} - 1 \right\}$	c ₆	$c_6 \sum_{j=2}^n \frac{j-1}{2}$
L7	Σ (n-1)	c ₇	c ₇ (n-1)

$$\text{Total running time, } T(n) = c_1 n + c_2(n-1) + \dots + c_7(n-1)$$

$$= O(n^2)$$